

Code:-

-- 1. Create the base tables

```
CREATE TABLE employees(  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    salary DECIMAL(10,2)  
);
```

```
CREATE TABLE employee_audit(  
    audit_id INT AUTO_INCREMENT PRIMARY KEY,  
    emp_id INT,  
    emp_name VARCHAR(50),  
    action_type VARCHAR(20),  
    action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

-- 2. Create AFTER INSERT Trigger

```
DELIMITER $$  
CREATE TRIGGER after_employee_insert  
AFTER INSERT ON employees  
FOR EACH ROW  
BEGIN  
    INSERT INTO employee_audit  
    (emp_id, emp_name, action_type)  
    VALUES  
    (NEW.emp_id, NEW.emp_name, 'INSERT');  
END $$  
DELIMITER ;
```

-- 3. Create AFTER DELETE Trigger

```
DELIMITER $$  
CREATE TRIGGER after_employee_delete  
AFTER DELETE ON employees  
FOR EACH ROW  
BEGIN  
    INSERT INTO employee_audit  
    (emp_id, emp_name, action_type)  
    VALUES  
    (OLD.emp_id, OLD.emp_name, 'DELETE');  
END $$  
DELIMITER ;
```

-- 4. Create AFTER UPDATE Trigger

```
DELIMITER $$  
CREATE TRIGGER after_employee_update  
AFTER UPDATE ON employees  
FOR EACH ROW
```

```

BEGIN
    INSERT INTO employee_audit
    (emp_id, emp_name, action_type)
    VALUES
    (NEW.emp_id, NEW.emp_name, 'UPDATE');
END $$
DELIMITER ;

-- 5. Create BEFORE INSERT Trigger (Validation)
DELIMITER $$
CREATE TRIGGER check_salary
BEFORE INSERT ON employees
FOR EACH ROW
BEGIN
    IF NEW.salary < 20000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary cannot be less than 20000';
    END IF;
END $$
DELIMITER ;

-- 6. Create BEFORE UPDATE Trigger (Validation)
DELIMITER $$
CREATE TRIGGER prevent_salary_reduction
BEFORE UPDATE ON employees
FOR EACH ROW
BEGIN
    IF NEW.salary < OLD.salary THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary reduction is not allowed';
    END IF;
END $$
DELIMITER ;

-- Test 1: This will succeed and automatically log to employee_audit
INSERT INTO employees VALUES (109, 'Arjun', 1, 52000);
SELECT * FROM employee_audit;

-- Test 2: This will succeed and log a DELETE action
DELETE FROM employees WHERE emp_id = 106;
SELECT * FROM employee_audit;

-- Test 3: This will fail and throw an error (Salary < 20000)
INSERT INTO employees VALUES (107, 'Ravi', 1, 15000);

-- Test 4: This will succeed (Salary > 20000)
INSERT INTO employees VALUES (108, 'Meena', 1, 30000);

-- (Setup for Test 5: Inserting employee 101 to test updates)

```

INSERT INTO employees VALUES (101, 'John', 1, 50000);

-- Test 5: This will fail (Salary reduction not allowed)

UPDATE employees SET salary = 40000 WHERE emp_id = 101;

-- Test 6: This will succeed and log an UPDATE action

UPDATE employees SET salary = 60000 WHERE emp_id = 101;

SELECT * FROM employee_audit;

-- Test 1: This will succeed and automatically log to employee_audit

Action Output				
	Time	Action	Response	Duration / Fetch Time
✓ 31	08:55:27	INSERT INTO employees VALUES (109, 'Arjun', 1, 52000)	1 row(s) affected	0.00100 sec

-- Test 2: This will succeed and log a DELETE action

audit_id	emp_id	emp_name	action_ty...	action_time
1	106	Arjun	INSERT	2026-08-22 08:50:04
2	106	Arjun	DELETE	2026-08-22 08:50:04
3	108	Meena	INSERT	2026-08-22 08:50:04
4	101	John	INSERT	2026-08-22 08:50:04
5	101	John	UPDATE	2026-08-22 08:50:04
6	106	Arjun	INSERT	2026-08-22 08:54:07
7	109	Arjun	INSERT	2026-08-22 08:55:27
NULL	NULL	NULL	NULL	NULL

-- Test 3: This will fail and throw an error (Salary < 20000)

Action Output				
	Time	Action	Response	Duration / Fetch Time
✗ 35	08:57:56	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be less than 20000	0.00034 sec

-- Test 4: This will succeed (Salary > 20000)

Action Output				
	Time	Action	Response	Duration / Fetch Time
✓ 41	09:00:14	INSERT INTO employees VALUES (110, 'Meena', 1, 30000)	1 row(s) affected	0.0026 sec

-- Test 5: This will fail (Salary reduction not allowed)

✗ 44	09:00:43	UPDATE employees SET salary = 40000 WHERE emp_id = 101	Error Code: 1644. Salary reduction is not allowed	0.00044 sec
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-- Test 6: This will succeed and log an UPDATE action

audit_id	emp_id	emp_name	action_ty...	action_time
1	106	Arjun	INSERT	2026-08-22 08:50:04
2	106	Arjun	DELETE	2026-08-22 08:50:04
3	108	Meena	INSERT	2026-08-22 08:50:04
4	101	John	INSERT	2026-08-22 08:50:04
5	101	John	UPDATE	2026-08-22 08:50:04
6	106	Arjun	INSERT	2026-08-22 08:54:07
7	109	Arjun	INSERT	2026-08-22 08:55:27
8	110	Meena	INSERT	2026-08-22 09:00:14
NULL	NULL	NULL	NULL	NULL